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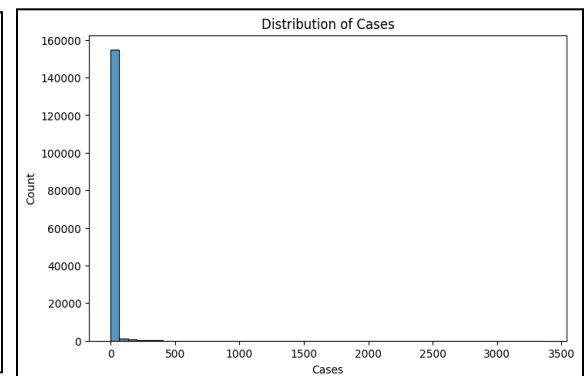
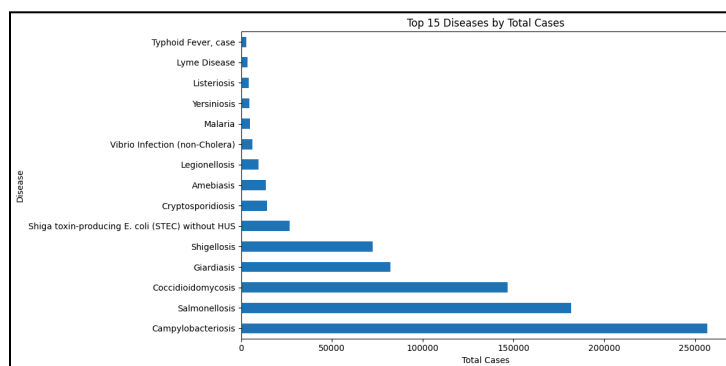
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Infectious Diseases Pattern Across California Counties

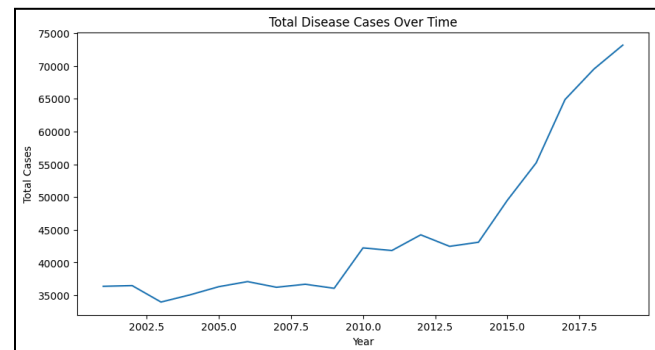
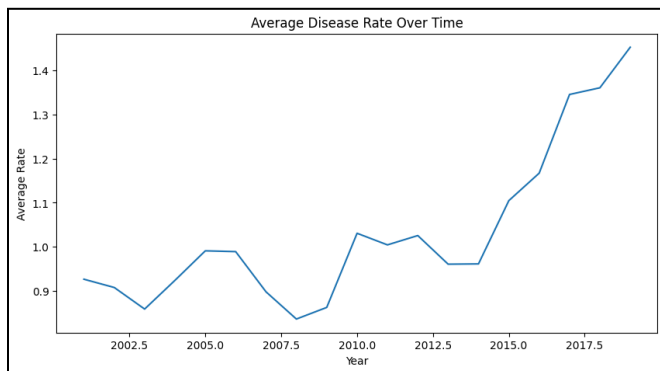
California's size, population diversity, and geographic variation make it an ideal setting for examining infectious diseases and various patterns at a large scale. Since 2001, the state's population has grown from approximately 34.5 million to 39.5 million residents, spanning over 58 counties, some more populous than others and others more contiguous, with living conditions shaped by external factors such as socioeconomic status, climatic zones, and racial groups. This report will investigate these patterns in infectious disease incidence across California from 2001 to 2019, while examining differences in incidence across counties. The findings are intended to inform the development of target public health prevention and intervention strategies tailored to the population and region.

The primary dataset for this project is the California Infectious Diseases Branch (IDB) dataset from Kaggle (<https://www.kaggle.com/code/kerneler/starter-infectious-diseases-county-dede22d4-b>), which not only contains reported infectious disease case counts across California counties from 2001–2019 but also includes variables relevant to public health analysis: 'Disease,' 'County,' 'Year,' 'Sex,' 'Cases,' 'Population,' 'Rate,' 'Lower_95__CI,' and 'Upper_95__CI.' The format for the variables is Disease, County, and Sex = string; Year, Cases, and Population = integer; Rate, Lower_95__CI, and Upper_95__CI = decimal.

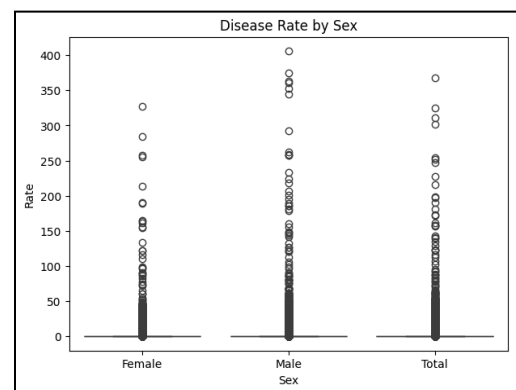
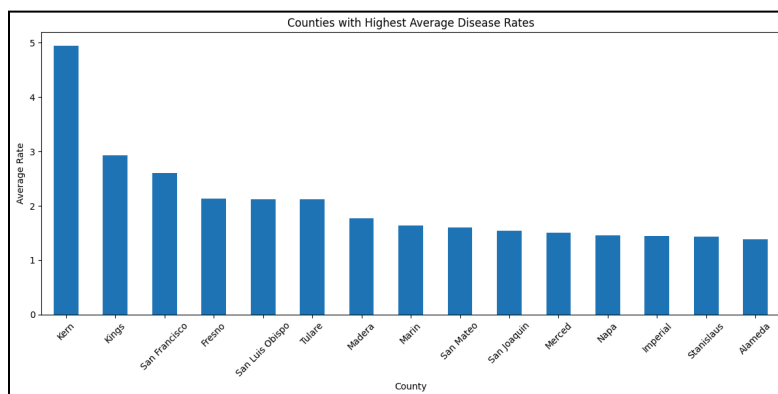
Prior to analysis, several data-cleaning steps were applied to ensure the dataset was ready for analysis. The other disease group was removed to ensure we had a clear idea of all the specific diseases mentioned, and because others could be used for multiple diseases, which could skew the data. Additionally, the rate column had “*” and had lots of values missing, so we decided to recalculate the rate using the rate data by multiplying the cases * 100000 / population, which fixed the issue of missing values in the data. There were some counties called California, which is the state, so all rows that were part of California were removed to ensure the analysis was conducted at the county level rather than at the state level. Additionally, an overview of the data was conducted to ensure no values were missing, and all unique diseases from 2001 to 2019 were present.



After the cleaning was done, we moved on to understand the data trends. We looked into the total cases grouped by disease and plotted them as a bar chart for the top 15 disease ranks. Then the distribution of raw case counts was examined using a histogram, revealing a strong right skew. We additionally plotted a log-transformed version ($\log(1 + \text{Cases})$) to better visualize the spread. A boxplot of disease rates was also used to identify outliers. We looked at trends by combining the total case count each year with the average disease rate across all years. For both, we saw a progressive upward trend, indicating that rates and cases increased over the years.

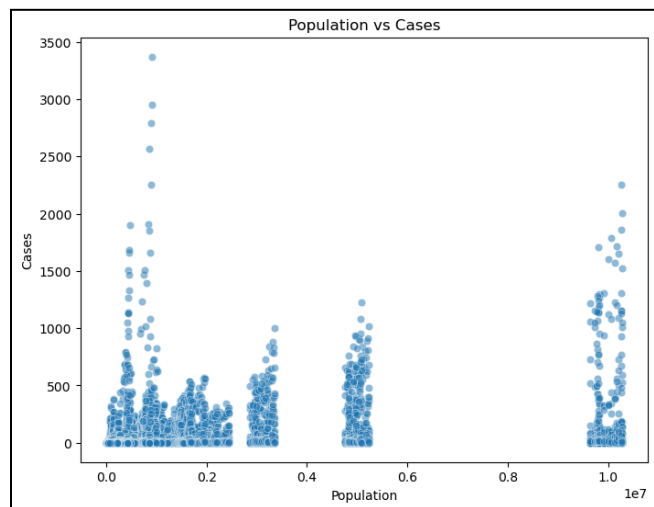


Additionally, we looked at geographic variation by looking at how average disease ratings were computed for the top 15 counties using a bar chart; the top 15 were Kern, Kings, San Francisco, Fresno, San Luis Obispo, Tulare, Madera, Marin, San Mateo, San Joaquin, Merced, Napa, Imperial, Stanislaus, and Alameda. Based on the graph, several of the highest-rate counties (Kern, Kings, Fresno, Tulare, Merced, and Madera) are concentrated in the Central Valley, suggesting a regional pattern that warrants further investigation and could enable more intervention and prevention in that area. Additionally, we looked into gender variations using box plots. This plot suggested that although the burden appears similar between the sexes, males experience more extreme high rates for certain diseases, motivating the formal chi-square and regression tests performed later.

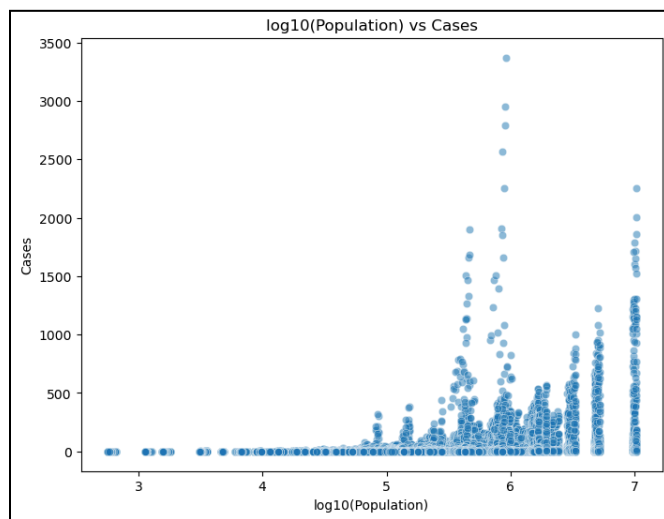


Next, we discovered a positive relationship between county population size and the number of disease cases. There are several distinct clusters: a few groups at very low population levels, one at the middle, and

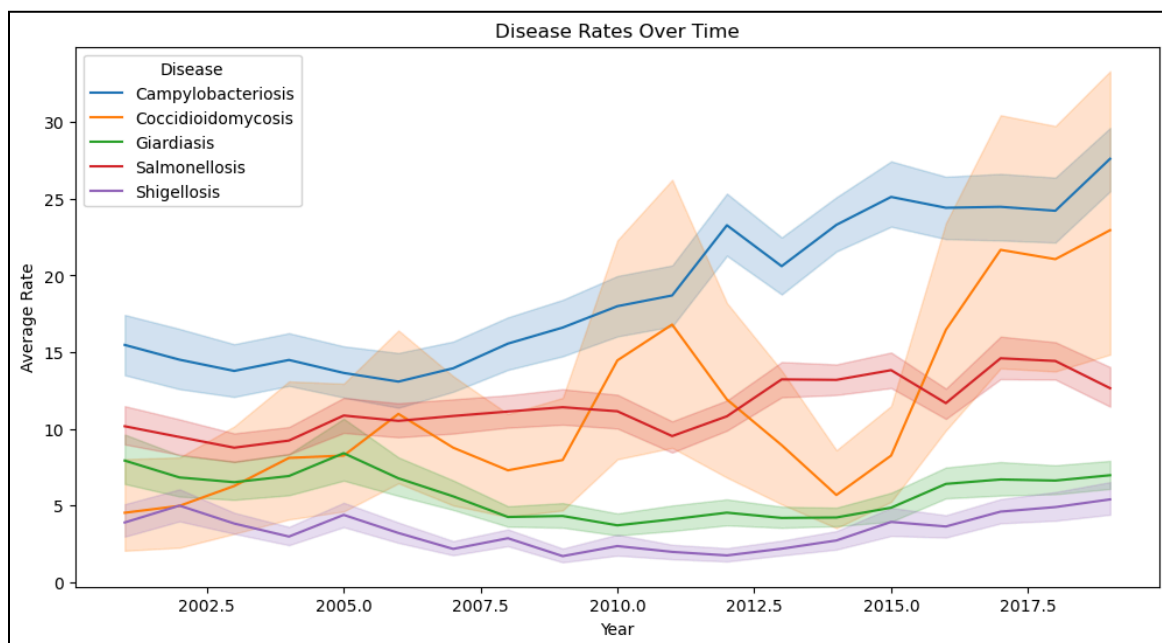
another cluster within the most populous counties. This suggests that California counties differ substantially in population size, creating natural groupings in the data. To better understand this relationship, we also examined a scatterplot using the logarithm of population. The log-transformed plot compresses the scale of the largest counties. The distribution is left-skewed, indicating that a relatively small number of highly populated counties account for a large share of the observations with high case counts. Both of these plots suggest that population is likely an important predictor of disease burden. Counties with larger populations naturally contain more individuals who may be exposed to infectious diseases, leading to higher case counts.



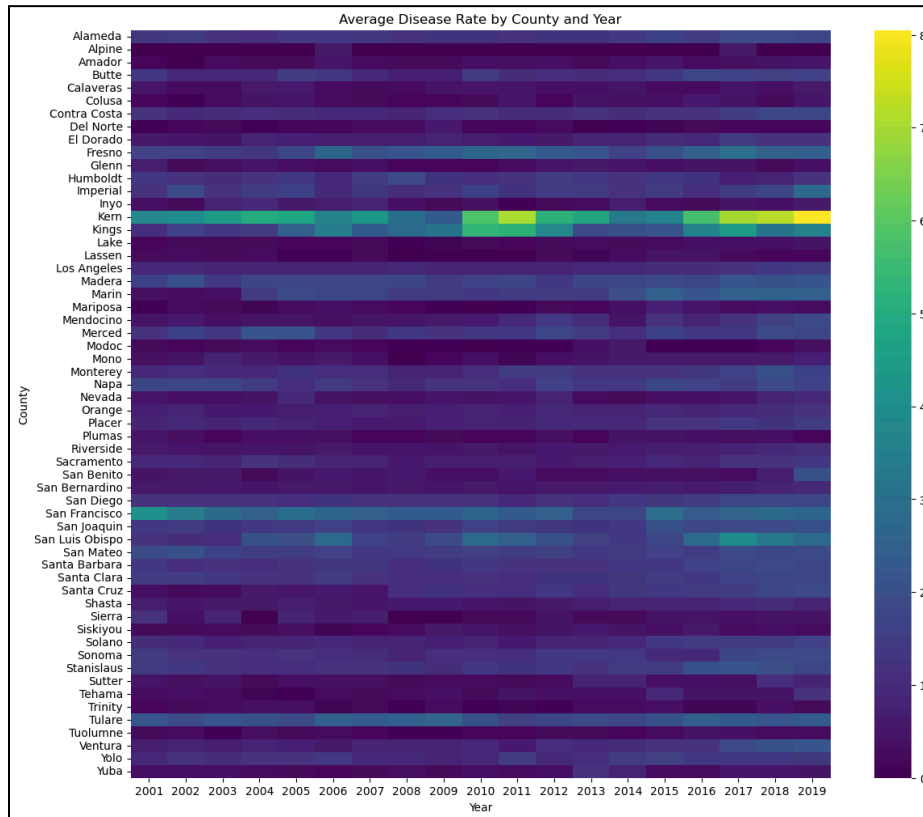
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temporal trends, we plotted the average incidence rate over time for the five diseases with the highest total number of reported cases. Giardiasis and Shigellosis diseases stayed relatively stable throughout the study period, showing little year-to-year variation in incidence rates. Another disease exhibits substantial fluctuations, with noticeable increases and decreases across different years. A fourth disease has a slight upward trend toward the end of the observation period, while a fifth disease shows a more moderate increase over time. These differing patterns indicate that disease incidence is not constant across years and that temporal effects may vary by disease. The diseases with relatively flat trajectories suggest stable transmission dynamics or reporting practices, while the more variable diseases may reflect outbreaks, changes in surveillance, environmental factors, or shifts in public health interventions.



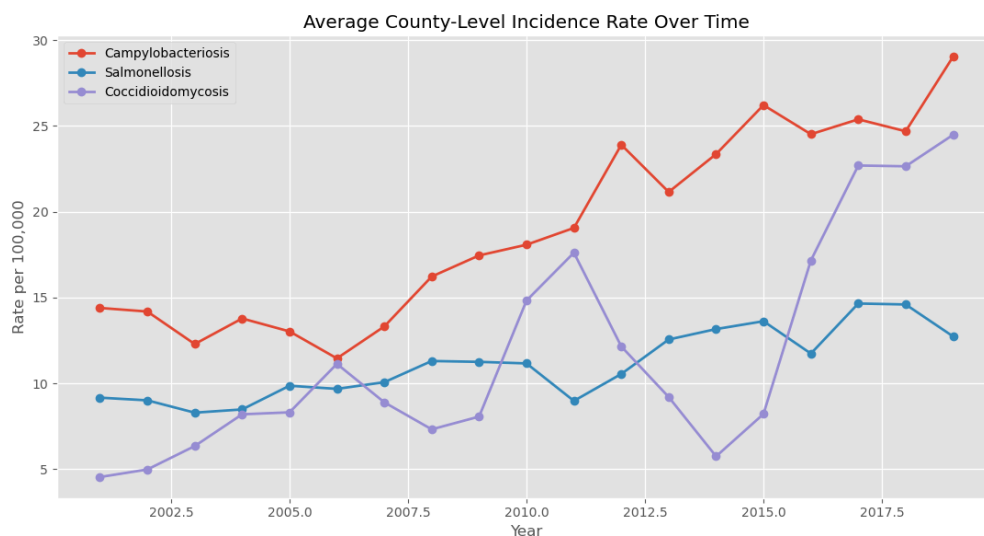
Finally, the county-year heatmap displays the average disease rate for each county across the study period. Most counties are in darker blue and purple shades, which mean relatively low average disease rates. However, several counties stand out with noticeably brighter colors. In particular, Kern County frequently appears in cyan and green shades, while Kings County is often represented by lighter blue and cyan tones, which suggests consistently higher disease rates compared to the other counties. Disease incidence is generally low for a large portion of the state, but a few elevated rates might mean there is some meaningful geographic heterogeneity in disease burden. Infectious disease incidence seems to be concentrated in specific regions.

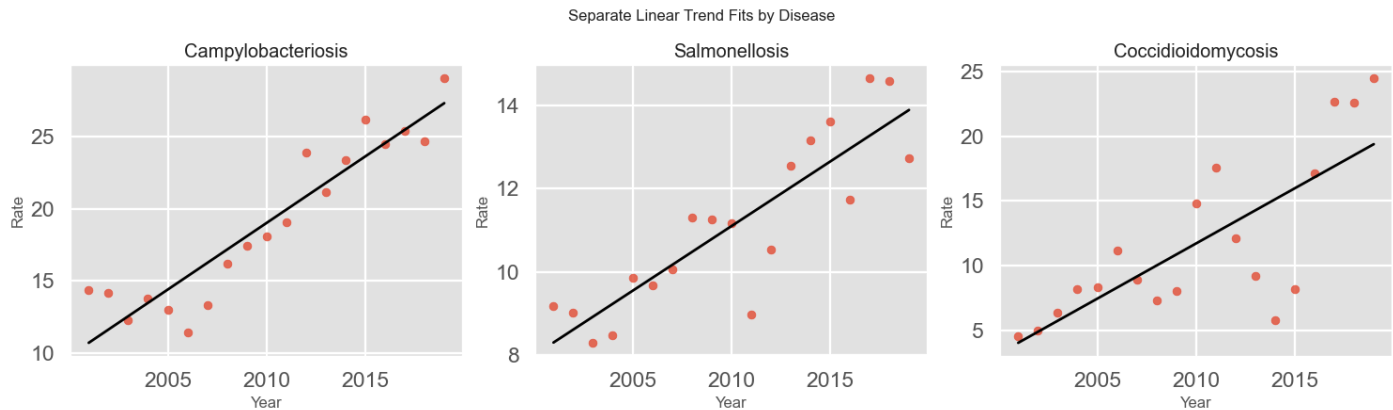


The primary objective of this study was to investigate whether infectious disease incidence varies across California counties and demographic groups, and to identify factors that may contribute to disparities under this pattern. Fortunately, the results from both the exploratory data analysis and the statistical analyses examined that infectious disease incidence is not uniformly distributed throughout California, and varies across both geographic regions and demographic groups. Before interpreting the results in detail, the analysis was organized so that each research question could be addressed with an appropriate statistical method. After exploratory visualizations to identify broad patterns in the data, the project then proceeded to regression modeling, nonparametric tests, and categorical association tests.

To examine temporal patterns in infectious disease incidence, we focused on *Campylobacteriosis*, *Salmonellosis*, and *Coccidioidomycosis*. These three diseases were selected because they contributed the largest number of cases in the filtered county-level dataset and provided enough repeated observations across years to

support regression analysis. Restricting the analysis to the most common diseases also reduced the influence of sparse diseases with many zero-count observations, which could otherwise obscure the main temporal patterns. Exploratory visualizations, including time-series plots and scatterplots with fitted regression lines, were first used to assess potential changes in incidence rates over time. After these visual summaries were examined, separate simple linear regression models were fit for each of the three selected diseases, with incidence rate as the response variable and year as the explanatory variable. This step allowed the rate of yearly change to be estimated for each disease individually and provided a formal test of whether the apparent temporal increases were statistically significant. A consistent upward trend across all three diseases between 2002 and 2017 was observed, which were also confirmed by linear regression of incidence rate. Based on highly statistically significant trends, Campylobacteriosis exhibited the strongest temporal increase, with incidence rates rising by approximately 0.92 units per year, followed closely by Coccidioidomycosis (0.85 units per year), while Salmonellosis increased at a more moderate rate of 0.31 units per year. Thus, disease incidence did not tend to increase constantly between 2001 and 2019, but reflected a significant increase for each of the major diseases examined. Even if the selected three diseases exhibited increasing incidence rates, the increasing magnitude was different, reflecting some diseases might experience faster growth over the study period. For example, Campylobacteriosis and Coccidioidomycosis demonstrated substantially steeper slopes than Salmonellosis.

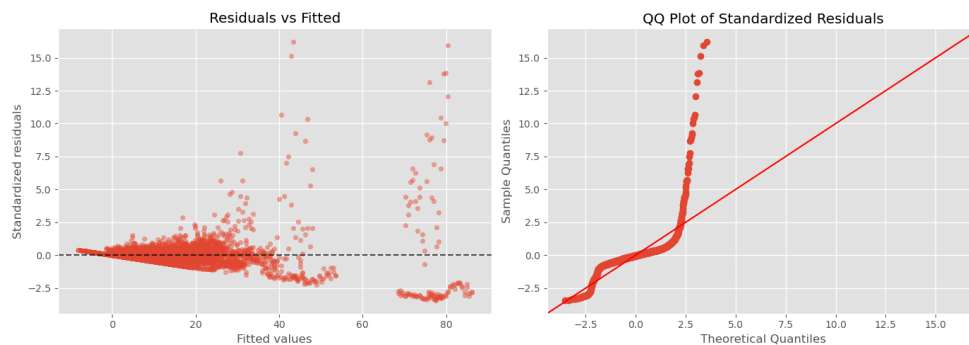




To determine whether diseases changed at similar rates over time, a regression model including a Year $\times$ Disease interaction term was fit. Since the separate regression results suggested that the slopes were not identical across diseases, an additional model was needed to test that difference directly. Rather than assuming one common time effect for all diseases, the next step was to compare a simpler model with a shared year effect to a more flexible model that allowed disease-specific time trends. This model allowed the effect of time to vary by disease rather than assuming a common temporal trend across all diseases. The interaction model provided a significantly better fit than the model containing only a common year effect ($p < 0.001$), indicating that temporal trends differed across diseases. Although all three diseases increased over time, Campylobacteriosis and Coccidioidomycosis experienced substantially steeper increases than Salmonellosis.

To evaluate whether temporal patterns persisted after accounting for geographic and disease-specific variation, a regression model was fit with year as the primary predictor while controlling for county and disease. After examining disease-specific trends, the analysis moved to a broader county-year regression framework. In this model, county and disease were included as control variables so that the effect of time could be evaluated independently of baseline differences across locations and disease categories. This made it possible to determine whether the increase over time remained even after accounting for persistent geographic and disease-specific variation. However, a highly significant predictor of incidence rate remained ($\beta = 0.566$, $p < 0.001$), which indicated that the increased incidence was not solely attributed to differences among counties or diseases. The estimated coefficient suggests that, after controlling for county and disease, the average incidence rate increased by approximately 0.57 units per year between 2001 and 2019. Thus, infectious disease burden increased between 2001 and 2019 with a different increasing rate across diseases. As a result, infectious disease incidence increased over the study period while also evolving differently across diseases. The significant overall year effect reflects a broad temporal increase in disease burden, whereas the significant interaction effect demonstrates that the magnitude of this increase was not uniform across all diseases. To assess model adequacy, residual-versus-fitted plots, a QQ plot of standardized residuals, and a Shapiro-Wilk normality test were also examined. These diagnostics indicated that the regression assumptions were not fully satisfied. The residual plot suggested heteroscedasticity and possible model misspecification, and both the QQ plot and Shapiro-Wilk test showed substantial non-normality in the residuals ($W = 0.4136$, $p = 2.844 \times 10^{-37}$). Therefore, the model remains

useful for describing overall time trends and differences across diseases, but formal statistical inference from the regression should be interpreted cautiously.



Because several parts of the dataset showed strong skewness and extreme values, not all comparisons were suitable for standard parametric procedures. For that reason, nonparametric methods were used for the county and sex comparisons so that the analyses would better match the shape of the observed distributions. In this setting, the Kruskal-Wallis test served as a nonparametric alternative to one-way ANOVA and was more appropriate for comparing many counties when the normality assumption was not reasonable. Thus, the test was selected to evaluate geographic differences without relying on overly strong distributional assumptions. The Kruskal-Wallis test produced an H statistic of 19,220.25 with a p-value less than 0.001. Therefore, we reject the null hypothesis that disease rates have the same distribution across all California counties, which also provides strong statistical evidence that infectious disease burden varies geographically throughout the state. The statistical findings were consistent with the patterns observed during the EDA. County-level summaries revealed substantial differences in average disease incidence rates, with Kern County exhibiting the highest average disease burden (mean rate = 4.93). Other counties with consistently elevated disease rates included Kings County and San Francisco County. In contrast, many counties exhibited substantially lower average incidence rates. As a result, these findings suggest infectious disease incidence varies substantially across California counties and that certain counties bear a considerably greater burden than others. This county-level analysis also complemented the earlier regression framework by showing that geographic variation was a relevant control variable and also an important feature of the dataset. Thus, the county comparisons helped clarify why it was necessary to account for location when modeling temporal disease trends.

A similar strategy was used to study demographic differences by sex. Since the distributions of disease rates for males and females were also skewed rather than approximately normal, a nonparametric two-group comparison was more suitable than a standard two-sample t-test. For this reason, the Mann-Whitney U test was chosen to compare disease-rate distributions between males and females. The Mann-Whitney U test came up with a highly significant result since the $p < 0.001$, which indicates that the distribution of disease rates differed between males and females. In addition to comparing the magnitude of disease rates, the analysis also considered whether disease occurrence could be viewed in a simpler categorical way. To do this, observations were grouped according to whether the disease incidence rate was positive or zero, and a chi-square test of independence was then used to examine whether this

occurrence pattern was associated with sex. Besides, a Chi-square test for independence was applied to examine whether disease occurrence itself was associated with sex. With the result of $\chi^2 = 24.86$, $p < 0.001$, we can conclude that there is a statistically significant association between sex and disease occurrence. In this case, males and females were not equally likely to experience positive disease incidence within the dataset. Sex was also examined within the regression framework rather than only through separate group comparisons. After fitting the main county-year model, sex was added as an additional categorical predictor in order to test whether it explained additional variation in incidence rate beyond year, county, and disease type. Finally, a fuller regression model combining sex, county controls, disease controls, and disease-specific time slopes was fit to evaluate whether sex still improved model fit when temporal differences across diseases were also allowed. The most important is, model performance improved significantly after sex was incorporated into the regression model. The addition of sex reduced the AIC and increased the model's explanatory power, indicating that sex contributes meaningful information beyond year, county, and disease type alone.

Although no prior published work was identified using this exact California county-year-sex dataset in the same way as this project, similar infectious-disease surveillance datasets are routinely analyzed by public-health agencies. For example, the California Department of Public Health's Epidemiologic Summaries describe infectious-disease incidence and distribution by county or region, sex, age group, race/ethnicity, and month of illness onset (California Department of Public Health, n.d.). The CDC's National Notifiable Diseases Surveillance System similarly supports analyses of case counts, rates, temporal trends, geographic patterns, and demographic differences (Centers for Disease Control and Prevention, 2024), and CDC dashboards are used to monitor case counts and rates over time filtered by demographics and geography (Centers for Disease Control and Prevention, n.d.). These examples show that the project's focus on time trends, county differences, sex differences, and disease burden is consistent with standard public-health surveillance analysis, even though the specific modeling approach was developed independently for this report.

Although the analyses revealed several significant patterns, this study has some limitations. First, the dataset was highly skewed and contained a lot of outliers, which required the use of nonparametric methods for several comparisons. Second, the demographic information available was limited to sex, so other potential important factors such as age, race, ethnicity, income, and healthcare access were not able to be analyzed. In addition, the data was aggregated at the county level, which may mask variation within individual counties and limit interpretation at the individual level. Another limitation is that the detailed temporal analysis focused only on the three diseases with the highest case counts. While this provided more stable estimates of long-term trends, the results may not generalize to less common diseases. Future work could incorporate additional demographic variables, spatial analysis techniques, and more advanced statistical models to better understand the factors driving geographic differences in disease burden.

This study investigated whether infectious disease incidence varied across California counties, demographic groups, and time between 2001 and 2019. The results showed that infectious disease burden is not uniformly distributed throughout California. Significant geographic differences were

observed across counties, with Kern and Kings counties showing some of the highest average incidence rates. Moreover, disease incidence increased significantly over time, especially for *Campylobacteriosis* and *Coccidioidomycosis*. The analysis also found statistically significant differences between males and females, and incorporating sex into the regression models improved overall model performance. Together, these findings suggest that infectious disease burden is influenced by geographic, demographic, and temporal factors. Understanding these patterns can help inform future public health surveillance efforts and support more targeted disease prevention strategies.

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